Human-centric Computing and Information Sciences

May 2026 | Volume 16



KCIA
Korea Computer Industry Association

www.hcisjournai.com

HCIS | Human-centric Computing and Information Sciences

RESEARCH (Research Manuscript) Open Access

Human-centric Computing and Information Sciences (2026) 16:27
DOI: https://doi.org/10.22967/HCIS.2026.16.027
Received: April 7, 2025; Accepted: June 21, 2025; Published: May 15, 2026

The Manifestation of Virtual Body Ownership in the Metaverse: A Cross-Platform Study on the Impact of Multisensory Integration on Presence and Flow Experience

Pengfei Li[1,*], Neal Xiong[2], and Jialin Li[1]

Abstract

This study explores the impact of virtual body ownership on presence and immersion, and examines the manner in which these factors influence the flow experience. Additionally, it examines the differences between virtual reality (VR) and smartphone platforms across the four aforementioned dimensions. Using experimental methods and questionnaire surveys, data were collected from 101 participants on university campuses. SPSS and AMOS tools, including correlation analysis, regression analysis, structural equation modeling, and paired-sample Wilcoxon signed-rank tests, were used to analyze data. The regression analysis results indicated that virtual embodied cognition had a significant positive impact on presence and immersion, whereas presence and immersion significantly influenced flow. The VR platform scored significantly higher than the smartphone platform in the dimensions of virtual embodied cognition and presence. However, there were no significant differences in the immersion and flow dimensions. These findings highlight the ability of VR to deliver superior embodied cognition and presence, thus enhancing its potential for immersive gameplay experiences.

Keywords

Virtual Body Ownership (VBO), Multisensory Integration, Experience of Presence, Experience of Immersion, State of Flow, Virtual Reality Platform, Smartphone Platform

1. Introduction

1.1 Background

With the rapid advancement of virtual reality (VR) technology, human–computer interaction is undergoing a fundamental transformation [1]. Compared with traditional video-based platforms, the defining characteristic of VR lies in the deep involvement of the user's body within the virtual environment [2]. By tracking head, hand, and limb movements, VR enables a closed-loop interaction between sensory input and motor output, allowing users not only to "see" themselves but also to "exist bodily" within the

※ This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (http://creativecommons.org/licenses/by-nc/3.0/) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.
*Corresponding Author: Pengfei Li (tlipengfei@jnu.edu.cn)
[1]College of Arts, Jinan University, Guangzhou, China
[2]Department of Computer, Mathematical and Physical Sciences, Sul Ross State University, Alpine, TX, USA

virtual space. This embodied experience overcomes the limitations of traditional media, which often position the body outside the screen, thereby enhancing both immersion and presence [3, 4].

Presence is commonly defined as the psychological state of "being there"—in which users temporarily suspend awareness of the real world and perceive the virtual environment as their immediate reality [5]. Immersion, on the other hand, refers to the degree of perceptual and attentional absorption enabled by the media, reflecting the extent of sensory coverage and user engagement [6]. Presence and immersion are closely related but represent distinct dimensions: subjective experience and technological affordance, respectively [7, 8]. Under conditions of high immersion and realism, users may enter a state of flow—a deep psychological state characterized by intense focus, enjoyment, diminished self-consciousness, and intrinsic motivation—wherein the activity is rewarding in itself [9].

Previous studies have shown that, compared to traditional media formats, VR is more effective at eliciting a sense of presence, emotional resonance, and flow experiences [10]. In VR systems, a user's real body can be substituted with a virtual body that matches their size and spatial orientation, and interactions can be experienced from a first-person perspective [11]. This mechanism, known as virtual embodiment, can induce the illusion of virtual body ownership (VBO), a psychological phenomenon in which individuals perceive a virtual body as part of their own body schema [12].

Psychological research suggests that the emergence of this illusion relies on the integration of multimodal sensory inputs, including visual, tactile, and vestibular cues, which, in turn, shape body schema and bodily self-consciousness [13]. When users identify with their virtual body, they not only experience a stronger sense of presence, but also deepen their immersion and enhance their flow state [14]. By contrast, user interaction in traditional video games is primarily mediated through external input devices such as game controllers, keyboards, or touchscreens. The body is only partially engaged, and its interactions with the virtual environment remain indirect [15]. In such contexts, immersion arises more from cognitive concentration and imaginative engagement than from full-body involvement [7].

VR systems enrich sensory experiences across visual, auditory, and haptic dimensions, enabling perceptual alignment between the virtual and physical bodies [16]. This synchronization facilitates a stronger sense of embodied presence within the virtual space, leading to more compelling and immersive experiences [17].

From the perspectives of philosophy and cognitive science, embodiment refers to the view that cognitive processes are not only confined to the brain, but also emerge through the dynamic interplay among the body, environment, and perceptual–motor activity [18]. Phenomenologist Maurice Merleau-Ponty has emphasized that the body is the origin of one's experiential world. According to his concepts of the "body schema" and "bodily intentionality," the body and consciousness are not separate entities but are intrinsically unified through perception and action [19]. In virtual environments, this theoretical framework provides a novel technological foundation: virtual embodiment integrates perception, action, and the environment through system-level interaction, thereby establishing a deeply symbiotic relationship between the user and the virtual world [20].

Neuroscientific research has revealed that virtual embodiments can activate the somatosensory cortex, motor cortex, and mirror neuron systems, thereby altering users' spatial perception and self-representation [21]. For instance, multimodal sensory synchrony not only enhances the sense of ownership over the virtual body but also facilitates the neural reconstruction of the "self-body" schema in virtual space, consequently inducing stronger senses of presence and flow experience [22]. Moreover, feedback from the virtual body engages psychological mechanisms related to emotional regulation and cognitive control, offering interdisciplinary pathways to understanding immersive experiences.

Although the Flow Theory, originally proposed by Mihaly Csikszentmihalyi, has been widely applied across domains such as education, gaming, and organizational management [23], its applicability within the context of multimodal embodied perception has not yet been systematically validated [24]. Existing studies conceptualize immersion and presence as antecedents of flow. However, in emerging media environments characterized by high levels of embodiment, traditional models fail to explain how virtual

body identification, mediated through multisensory integration, contributes to the emergence of flow states.

Accordingly, this study focuses on the core question of how multimodal virtual body perception influences users' embodied presence and flow states. We extend the classical flow model by introducing "virtual embodied cognition" (VEC) as a mediating construct. VEC emphasizes the dynamic synergy between multisensory input, virtual body identification, and perceptual engagement, aiming to uncover its impact on immersion, presence, and ultimately, flow.

Furthermore, to recognize the mediating effects of media modalities on user experience, this study employed a cross-platform experimental design (VR vs. smartphone). By taking the game content as a constant, we systematically compared how different media platforms influenced VEC, immersion, presence, and flow states. Through quantitative analysis and path modeling, this study seeks to expand the theoretical boundaries of embodied cognition within interactive media and offers new perspectives and empirical support for immersive experience design, VR narrative practice, and the study of the underlying psychological mechanisms

1.2 Research Hypotheses

1.2.1 Virtual embodied cognition

With the advancement of VR technology, researchers have increasingly recognized the critical role of the avatar's body in virtual environments and have focused on how users achieve embodied cognition within these spaces [25]. As phenomenologist Maurice Merleau-Ponty has argued, the body is the starting point and primary medium through which humans make sense of the world [26, 27]. Similarly, in virtual contexts, the virtual body serves as a fundamental coordinate axis for user cognition [28].

1.2.2 Virtual body and mechanisms of embodied cognition

In VR environments, users are required to perceptually integrate their physical body with a virtual avatar by controlling the avatar to perform interactive behaviors, thereby gradually establishing avatar identification [29]. Prior research has identified three core mechanisms that underpin VEC: activation of body schema, sense of body ownership, and alignment of motor intentions (agency) [30]. High-precision full-body tracking, particularly real-time mapping of the hands, feet, and torso, has been shown to significantly enhance users' sense of embodiment and spatial presence [31, 32].

1.2.3 Multisensory integration and embodied interaction

Recently, the theoretical boundaries of embodied cognition have been further expanded, with increasing emphasis on embodied interaction and multisensory integration as critical factors shaping the immersive experience. By manipulating visual perspectives and incorporating auditory and tactile feedback, researchers have successfully induced a body-swapping illusion, thereby elucidating the role of multisensory inputs in the formation of VBO [33]. Studies have shown that enhanced tactile feedback not only strengthens the sense of ownership over the virtual body, but also improves users' cognitive adaptation and immersion in virtual environments [34].

1.2.4 Immersive cognition and the flow experience

With research expanding from sensory inputs to cognitive mechanisms, immersive cognition has emerged as a key theoretical framework. This approach posits that, in highly immersive virtual environments, users undergo a dynamic reconfiguration of their cognitive boundaries regarding space, time, and self, making them more susceptible to experiencing flow states [35, 36]. The relationships among immersion, presence, and flow are not strictly linear; rather, these factors interact synergistically and mutually reinforce each other. Furthermore, neuroscientific studies have demonstrated that embodied experiences are not solely dependent on somatosensory processing but also involve brain regions associated with emotion regulation and attentional control [37]. These findings open new avenues for

understanding VEC at the neural level.

In summary, VEC is no longer understood merely as the perception of bodily control or motor feedback but rather as an integrative mechanism involving multisensory inputs and cognitive adaptation. Building on the previous literature, this study incorporates recent advances in embodied interaction, multisensory integration, and immersive cognition to deepen our understanding of avatar identification mechanisms and their impact on immersive experiences and flow states. Through this updated theoretical lens, our research aims to engage more effectively with the current academic discourse on VR-based human–computer interaction and virtual presence.

1.2.5 Experience of presence

Some scholars have differentiated presence or sense of being there from immersion. Research indicates that multisensory stimuli can enhance user immersion, whereas a virtual avatar's free movement and accurate reflection of user actions significantly bolster presence. Moreover, highly realistic virtual environments can simultaneously increase immersion and presence [38]. Concerning the relationship between virtual bodies and presence, high-fidelity virtual bodies can significantly enhance users' sense of social presence and the illusion of place [39]. Several technical factors are considered as crucial for enhancing presence such as stereoscopic vision, a wide field of view, and high interactivity [40]. Caroux and Mougine [41] highlight the significant influence of display modalities on the level of presence, with head-mounted displays (HMDs) offering a higher presence than traditional screens. Kim et al. [42] revealed that users' physical postures on a VR platform significantly affected presence, which was closely tied to the proprioception of the physical body and the embodied cognition of the virtual body. Haans and IJsselsteijn [43] conceptualized virtual presence on three levels: "morphology," involving the physical characteristics of the body such as the number, type, and positioning of limbs; "body schema," an unconscious system that regulates posture and movement, enabling smooth interaction with the environment; and "body image," the subjective perception and conceptual knowledge of one's body. Thus, embodied cognition achieved through physical body participation is closely linked to presence in virtual spaces. This sense of presence relies on media technology, allowing users to perceive the existence of virtual bodies through multisensory experiences of their physical bodies. It is a holistic experience determined by body characteristics, schemas, and images [44].

1.2.6 Experience of immersion

Although clearly distinguishing between presence and immersion is challenging in virtual-space experiences, their core emphases differ. Presence primarily relates to cognitive and psychological responses, referring to a user's feeling of "being in a virtual environment." In contrast, immersion is typically associated with sensory feedback, describing the extent to which users feel "enveloped by the virtual environment" [41]. From a Cartesian perspective, the body and consciousness are separate entities, with the body more closely associated with presence, and consciousness with immersion.

In contrast, Maurice Merleau-Ponty posited that the body and consciousness are inseparable, forming an integrated whole that transcends the sum of their parts. Thus, presence and immersion are overlapping manifestations of this holistic perception, representing the union of body and mind. Immersion resembles the bodily experience of "being submerged in water," surrounded by an entirely different reality, capturing the user's full attention and sensory perception and leading to a "physiological sense of being there" [45]. However, immersion is not solely derived from sensory stimuli. Compelling narratives can also influence immersion, such as reading a fascinating book [46]. The mechanisms for achieving immersion and presence differ. Immersion is commonly understood as a psychological state experienced during engagement, concentration, or deep absorption, encompassing sensory "envelopment" and a focused psychological experience [47]. Dooley explored the impact of embodiment and immersion on users' psychological states, indicating that higher levels of embodiment led to stronger emotional engagement and deeper immersion [48]. This highlights the interplay between physicality, attention, and narrative in shaping users' immersive experiences.

1.2.7 Flow state

Csikszentmihalyi [49] defines "flow" as an optimal and enjoyable state that individuals enter when fully immersed in an activity. In the state of flow, attention is highly focused, task engagement is profound, and individuals experience increased excitement and satisfaction. People in a flow state often exhibit feelings of relaxation and happiness, along with qualities such as "concentration, mental energy, and creativity." Flow theory is widely applied in game studies because games are considered ideal media for fostering flow experiences. Csikszentmihalyi [49] identified several defining elements of flow, including the "reduction or disappearance of worry." Flow experience describes an individual's behavior and perception during flow and is characterized by deep concentration, loss of self-awareness, a distorted sense of time, and a profound sense of presence. Flow outcomes refer to the positive emotional and behavioral changes resulting from flow, such as enhanced positive emotions and immersive goal-driven experiences. These theoretical frameworks provide insights into the flow mechanisms and offer valuable guidance for designing systems that optimize the user experience of presence and immersion. Designers can create environments that encourage users to achieve deeper engagement and satisfaction through understanding and fostering flow.

Flow is often described as an "extreme, optimal, and positive" experience [50, 51]. Presence is frequently used interchangeably with terms such as immersion, flow, and telepresence [50]; an experience becomes compelling to users when their cognitive state aligns with their preferences, resulting in immersion [52]. Presence is defined as the "psychological sensation of being in a virtual environment" [53]; this implies that the individual's cognitive and perceptual systems are "tricked" by the stimuli of the virtual environment, leading to a sense of being in the virtual space rather than their physical location [54]. Immersion occurs when an individual's cognitive state is captured by experience, and they develop a preference for virtual content. At this point, their perceptual state detaches from the physical world and transitions into the virtual realm, allowing them to experience presence in virtual space. This forms the foundation for the subsequent flow state. Therefore, immersion and presence serve as critical precursors to flow, linking the users' embodied cognition and engagement with a virtual environment.

On a VR platform, the sense of presence associated with a virtual body is cultivated through technical simulations of visual, auditory, and haptic stimuli. This approach allows users to share virtual bodies with their avatars by immersing themselves in virtual scenarios and narratives. During this process, the user's perceptual and cognitive systems become fully engaged with their body and mind, enabling them to enter a state of flow.

This progression relies on the sense of presence and immersion fostered by the virtual body, and is facilitated by embodied cognition and sensory stimuli, ultimately leading to a flow state. Based on this framework, we propose the following hypothesis:

Hypothesis 1: VEC positively impacts the experience of presence (EP).

Hypothesis 2: VEC positively impacts the experience of immersion (EI).

Hypothesis 3: EP positively impacts the flow state (FS).

Hypothesis 4: EI positively impacts FS.

Based on the above discussion, a hypothetical model for VEC was designed, as shown in Fig. 1.

1.2.8 Comparative studies on player experiences in VR and non-VR platform

Numerous studies have compared the player experiences on VR and non-VR platforms. The findings indicated that the VR platform score was significantly higher across most experiential dimensions than traditional non-VR platforms. However, VR platforms continue to face challenges, particularly in simulated vertigo [55]. Researchers have also compared gameplay experiences between VR systems and desktop displays. Despite the similar performance outcomes under both conditions, VR elicits stronger positive emotions [56] and significantly higher levels of immersion and flow [57]. Another study compared player experiences across different game genres, such as highly interactive strategy games and less interactive racing games, by using VR and traditional platforms. The results revealed that participants consistently experienced a stronger sense of presence when playing VR, regardless of the game type [58].

Scholars have extensively studied user experiences on VR and non-VR platforms, covering dimensions such as presence, immersion, and flow. These studies have further explored how these experiences relate to the behaviors and dynamics in virtual spaces. However, comparative research on embodied cognition in virtual spaces with user bodily engagement needs to be expanded, and the comparison of VEC with presence, immersion, and flow in VR and non-VR contexts must also be addressed [59]. Given this gap, our team conducted a comparative study based on a VR game and traditional video game platform. The same game was used across both platforms, which allowed us to observe the differences in user experiences of VEC, presence, immersion, and flow between the two platforms. Given the widespread popularity of smartphones as gaming media and user familiarity with smartphone platforms, our study focused on this factor. Based on this framework, we propose the following hypotheses:

Hypothesis 5: Users experience more significant embodied cognition on a VR platform than on a smartphone.

Hypothesis 6: Users experience more significant presence on a VR platform than on a smartphone.

Hypothesis 7: Users experience more significant immersion on a VR platform than on a smartphone.

Hypothesis 8: Users experience more significant flow on a VR platform than on a smartphone.

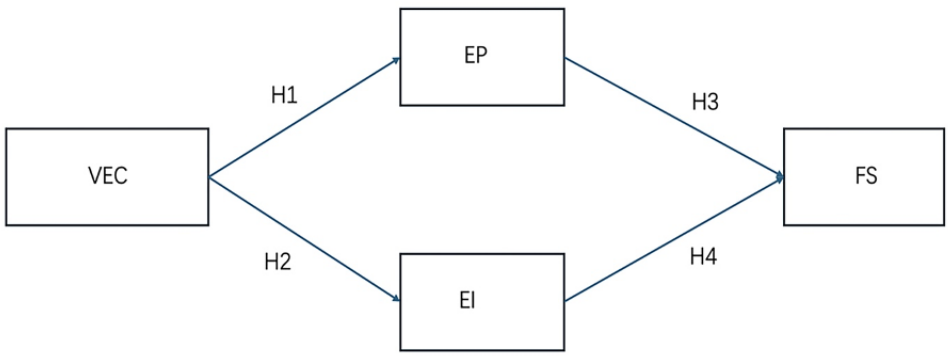


Fig. 1. The hypothesis framework of four dimensions and incarnation of virtual embodied cognition (VEC) hypothesis model.

2. Materials and Methods

2.1 Design and Study Population

The research team recruited healthy adults from multiple universities in China. Participants were required to have a certain level of familiarity with gaming, adequate reading and verbal comprehension abilities, and no significant visual or auditory impairments. The VR platform utilized the Oculus Quest 2 device, whereas the smartphone platform employed a Huawei Mate 40. The experimental game selected was "Acron Attack of the Squirrels," a competitive game available on both VR and smartphone platforms supporting cross-platform gameplay.

To enhance the statistical power and representativeness of the study, an a priori power analysis was conducted using G*Power 3.1 software ($\alpha = 0.05$, effect size $f = 0.25$, power = 0.80), which indicated a minimum sample size of 72 participants. Ultimately, 101 participants were included, which met the sample size requirements. Moreover, to increase sample heterogeneity and external validity, participants were drawn not only from Jinan University, but also from other higher education institutions in the Guangzhou area, including Sun Yat-sen University and South China Normal University.

In addition, to improve the internal validity of the experiment, the following confounding variables were controlled.

- Field of view (FOV): All VR experiences were conducted using standard Oculus Quest 2 settings to ensure consistent visual exposure.
- Input modality: Interaction methods were fixed by device type (VR controllers for VR, touchscreen for smartphones) to exclude the influence of input variation on user experience.
- Spatial layout: All experiments were conducted in the same laboratory environment to maintain consistent background noise and spatial arrangements.

Owing to limitations in the experimental design, it was not feasible to consider all hardware media forms of traditional video games. Therefore, a representative medium was selected for comparison with VR gaming. The chosen game, Squirrel Battle, supports the competition between VR and smartphone platforms. The research team selected smartphones as the traditional video game medium for this study because compared to gamepads or mouse controls, the smartphone touchscreen offered ease of use and intuitive interaction.

2.2 Experimental Procedure

Pre-test: Participants completed a basic information questionnaire including age, gender, and gaming experience. The baseline test used the Flow State Scale (FSS) to measure the participants' initial flow experiences.

Game experience: The participants were randomly divided into two groups: one played the VR game first, and the other played the smartphone game first. Each participant underwent a 10-minute familiarization session with the equipment to ensure that they were comfortable with the controls. The participants were paired with one player on the VR platform and another on the smartphone platform for a 20-minute competitive gaming session.

Intermediate measurement: After the gaming session, the participants immediately completed questionnaires to assess their embodied cognition, presence, immersion, and flow state during the game. A short break of approximately 10 minutes was provided to prevent fatigue from affecting the results.

Cross-test: The two groups swapped gaming devices and underwent another 10-minute familiarization session with the new equipment. They played the other version of the game for 20 minutes. After this session, the participants completed questionnaires to assess their VEC, presence, immersion, and flow.

In this study, the FSS was employed during the pre-test phase to obtain a baseline measurement of participants' flow states. The first objective was to establish baseline establishment: prior to the experimental procedure, all participants completed the FSS to assess their current flow levels, ensuring that there were no significant differences in flow among participants at the outset. These baseline data served as a reference for subsequent comparisons across the experimental conditions. The second objective was to screen the participants. The FSS was also used to select participants who met specific criteria, ensuring that only those with relatively balanced flow levels were included in the experiment. This approach aimed to minimize the potential confounding effects of preexisting flow differences on the results. By adopting this two-stage pre-test procedure, the study enhanced its internal validity and ensured that comparisons between experimental conditions were both meaningful and analytically robust.

This experimental design ensures that both game versions are compared in a balanced and systematic manner while minimizing potential biases related to device familiarity and fatigue.

We randomly invited participants from various campuses at Jinan University to participate and paired them into groups of two. Each pair alternately completed gameplay experiences on the VR and smartphone platforms, followed by a questionnaire. Subsequently, the participants switched devices to repeat the experience, and completed a questionnaire. An electronic questionnaire was developed using the "Questionnaire Star" Platform. The entire experiment was conducted over a three-month period, from July 16, 2024 to October 20, 2024, and additional participants were recruited in June 2025. A total of 101 participants were recruited via WeChat Moments. All participants voluntarily agreed to participate after reading the informed consent form and allowed the collection of data in this survey, including mobile information, to be used by the research team or other researchers for further studies. Due to limited

research funding, the complexity of the experimental process, and the fact that all participants were adults, the sample size for this study was relatively small. Therefore, the data obtained from this survey may not fully represent the actual needs of all the users. However, based on these findings, the survey provides meaningful insights into the impact of VR platforms versus traditional video platforms on user experiences, and aids in understanding the relationships between VEC, EP, EI, and FS.

2.3 Measurement Tools

The measurement scales used in this study were established and validated using the Virtual Embodiment Questionnaire (VEQ) (12 items) [60], Experience of Presence Questionnaire (EPQ) (32 items) [61], Experience of Immersion Questionnaire (EIQ) (17 items) [62], and FSS (28 items) [63]. A 5-point Likert scale was used to score the items (1 = strongly disagree, 5 = strongly agree). To enhance readers' understanding of the structure of the VEQ, this study briefly presented sample items from each sub-dimension. For instance, in the Ownership dimension, a representative item was "I felt as if the virtual body were my own body." In the Agency dimension, an example item was "I felt that I could control the movements of the virtual body." In the Body Change dimension, a typical item was "I felt that my body had changed and become like the virtual character." All items were rated on a 5-point Likert scale, with higher scores indicating a stronger sense of virtual body perception.

2.4 Ethical Considerations

The present study was approved by the Institutional Review Board of the Social Sciences and Humanities Department of Jinan University, China (IRB No. A2407001-037). The original informed consent document stated that researchers or other non-team researchers could use the mobile information collected during the survey for subsequent studies. All data were anonymously processed to ensure confidentiality. Additionally, each participant received a VR game valued at $4.99 as compensation. Participant anonymity was fully safeguarded, and no individual participants could be identified from any images included in the manuscript or supplementary material.

3. Results

3.1 Data Collection

We collected two sets of valid questionnaires on the VR and smartphone platforms, both completed by 101 participants. All participants were fully informed and consented to providing information on the questionnaire. The response rate was 100%. Table 1 presents the descriptive statistics of the survey sample.

Table 1. Sample descriptive statistics (n=101)

Variable	Number of participants (%)
Sex	
Male	42 (41.6)
Female	58 (57.4)
Intersex	1 (1.0)
Age group	
18–23 years	68 (67.3)
24–29 years	31 (30.7)
30–35 years	0 (0)
≥36 years	2 (2.0)

Continued on the next page.

Table 1. Continued

Variable	Number of participants (%)
Career	
Student	99 (98.0)
Civil servant	0 (0)
Private-sector employee	0 (0)
Self-employed	0 (0)
Working in a medical or educational institution	1 (1.0)
Other	1 (1.0)
Educational background	
High school (or vocational school) and below	2 (2.0)
Bachelor's or associate degree	39 (38.6)
Master's degree and above	60 (59.4)
Average time spent gaming per week	
Almost no gaming	14 (13.9)
<1 hour	10 (9.9)
1–5 hours	29 (28.7)
5–10 hours	22 (21.8)
>10 hours	26 (25.7)
Played VR games before	
Yes	74 (73.3)
No	27 (26.7)

3.2 Analysis

The final questionnaire, which focused on the VR game experiences, underwent reliability and validity analyses using SPSS software version 27 (IBM Corp., Armonk, NY, USA). In the reliability analysis, the Cronbach's α coefficient for the overall scale was >0.8 (0.912), indicating excellent data reliability, as shown in Table 2. For validity, the data were subjected to the Kaiser-Meyer-Olkin (KMO) test and Bartlett's test of sphericity. The KMO statistic was >0.8, and the corresponding *p*-value was <0.001. These results suggest that the questionnaire exhibits strong overall validity and supports the factor analysis (Table 2).

Table 2. Cronbach's reliability analysis (n=101; overall Cronbach's α >0.8)

Variable	Number of items	Sample size	Cronbach's α
Virtual embodied cognition (VEC)	12	101 (100%)	0.931
Experience of presence (EP)	32	101 (100%)	0.970
Experience of immersion (EI)	17	101 (100%)	0.933
Flow state (FS)	28	101 (100%)	0.964

Textbox 1. Mapping of variables to factors (VEC=12, EP=28, EI=16, and FS=26)

Variables and factors

VEC: items AG (1,2,3,4), OW (1,2,3,4), CH (1,2,3,4),

EP: items 1,2,3,4,5,6,7,8,9,11,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29, and 31

EI: items 1,2,3,4,5,6,7,8,9,10,12,13,14,15,16,17

FS: items 1,2,3,4,5,6,7,8,9,10,11,12,14,15,16,17,18,19,21,22,23,24,25,26,27, and 28

After the initial review, exploratory factor analysis was conducted to assess the validity of the questionnaires. Of the four scales, three had two dimensions each, and the VEQ consisted of three dimensions. Items with factor loadings greater than 0.5 for each dimension were retained, while those that did not pass the validity check were removed. The items deleted included items #10, 12, 30, and 32 from EPQ; item #11 from EIQ; and items #13 and 20 from FSS. The remaining items were considered valid and were retained after the validity check, as shown in Textbox 1.

Based on a comprehensive analysis, the items from the four questionnaires were categorized as follows:

1) VEQ: Items AG1, AG2, AG3, AG4, OW1, OW2, OW3, OW4, CH1, CH2, CH3, and CH4.
2) EPQ: Items 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, and 31.
3) EIQ: Items 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, and 17.
4) FSS: Items 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 14, 15, 16, 17, 18, 19, 21, 22, 23, 24, 25, 26, 27, and 28.

The results indicate good structural validity of the data, and the correlation analysis results are presented in Table 3.

Table 3. Results of the correlation analysis of the average scores for the four dimensions in the model (n=101)

	VEC		EP		EI		FS	
	r	***p*-value**	***r***	***p*-value**	***r***	***p*-value**	***r***	***p*-value**
VEC	1	N/A	0.715	<0.001	0.737	<0.001	0.613	<0.001
EP	0.715	<0.001	1	N/A	0.820	<0.001	0.723	<0.001
EI	0.737	<0.001	0.820	<0.001	1	N/A	0.795	<0.001
FS	0.613	<0.001	0.723	<0.001	0.795	<0.001	1	N/A

N/A=not applicable.

The regression analysis results indicated that all independent variables had significant positive predictive effects on the dependent variables. The variance inflation factors (VIFs) for all predictors were below five, suggesting that no multicollinearity issues were present. Specifically, VEC significantly predicted presence ($\beta = 0.715$, $p < 0.001$), and had an even stronger predictive effect on immersion ($\beta = 0.737$, $p < 0.001$). Additionally, presence significantly predicted flow state ($\beta = 0.218$, $p = 0.039$), and immersion also showed a significant positive effect on flow ($\beta = 0.616$, $p < 0.001$). According to the standards proposed by Cohen [64], standardized β values between 0.3 and 0.5 indicate moderate effects, while values above 0.5 indicate strong effects. Therefore, in the current model, VEC exerts a strong influence on both presence and immersion, whereas the remaining path coefficients represent moderate effect sizes, as shown in Table 4.

Table 4. Results of regression analysis

Dependent variable	Independent variable	Unstandardized coefficients, B (SE; 95% CI)	Standardized coefficients, β	t-test (df)	*p*-value	Collinearity statistics	
						Tolerance	VIF
EP	VEC	15.222 (1.496; 12.253–18.191)	0.715	10.173 (100)	<0.001	1.000	1.000
EI	VEC	9.551 (0.881; 7.802–11.300)	0.737	10.836 (100)	<0.001	1.000	1.000
FS	EP	0.184 (0.088; 0.009–0.360)	0.218	2.088 (100)	0.039	0.328	3.048
	EI	0.854 (0.145; 0.566–1.142)	0.616	5.886 (100)	<0.001	0.328	3.048

VIF=variance inflation factor.

3.2.1 Structural equation model analysis

The correlation analysis results indicated significant relationships between VEC, experience of presence, experience of immersion, and flow state. To further validate the hypothesized relationships among the four variables, structural equation modeling (SEM) was conducted using SPSS version 27 and SPSS Amos version 25. The 95% confidence intervals for the estimates were assessed, and intervals that did not contain 0 were deemed statistically significant. The model fit indices were as follows: $\chi^2 = 82.780$, $\chi^2/df = 3.599$ (between 3.0 and 5.0, the model fit is considered moderate but acceptable), Incremental Fit Index (IFI) = 0.914(>0.9), Comparative Fit Index (CFI) = 0.913 (>0.9), Tucker-Lewis index (TLI) = 0.863 (>0.85), and Normalized Fit Index (NFI) = 0.885 (>0.85). Except for the CFI and NFI values, which were slightly below the recommended threshold of 0.9, all the other fit indices fell within the acceptable range, indicating that the theoretical model was acceptable. The results demonstrate the following significant relationships: VEC significantly impacts the experience of presence ($\beta = 0.091$, $p < 0.001$), VEC significantly impacts the experience of immersion ($\beta = 0.096$; $p < 0.001$), the experience of immersion does not significantly impact flow state ($\beta = 0.146$; $p = 0.229$), and the experience of presence significantly impacts flow state ($\beta = 0.156$; $p < 0.001$). The hypothesized relationships between VEC and presence, VEC and immersion, and presence and flow were supported and found to be statistically significant. However, the hypothesized relationship between immersion and the flow state was not significant and was therefore not supported, as shown in Table 5.

Table 5. Structural equation model results validation (hypothesis supported)

Pathway	Unstandardized coefficient estimate	Standardized coefficient estimate	Critical ratio	*p*-value	Hypothesis supported
H1: EP ← VEC	0.726	0.091	8.023	<0.001	Supported
H2: EI ← VEC	0.848	0.096	8.867	<0.001	Supported
H3: FS ← EI	0.175	0.146	1.203	0.229	Not supported
H4: FS ← EP	0.575	0.156	3.688	<0.001	Supported

Hypothesis verification and model analyses were conducted using graphical representations. The relationships among the four variables are shown in Fig. 2.

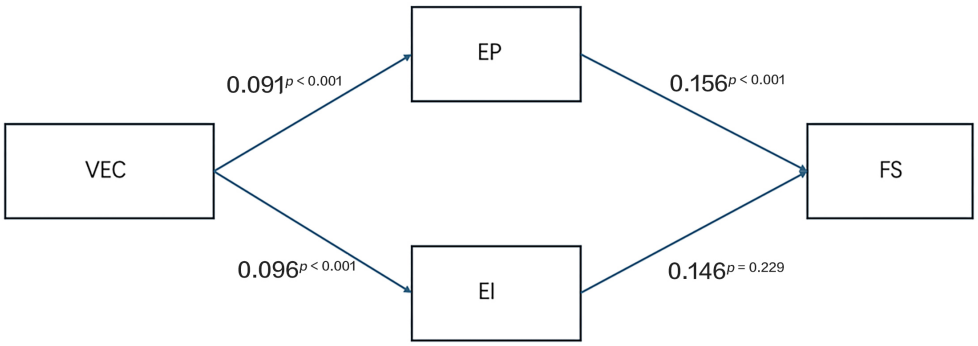


Fig. 2. The influence relationships among the four variables.

3.2.2 Comparison and analysis of differences between VR platform and smartphone platform

Questionnaire data from the comparative tests of the VR and smartphone platforms were analyzed using SPSS software version 27. After testing for normality, the results indicated non-normal data. The paired Wilcoxon signed-rank test was used to compare two groups. The following conclusions were drawn:

VEC: Users experienced significantly higher embodied cognition on the VR platform than on the smartphone platform ($Z = -2.277$, $p = 0.023$). The effect size was $r = -0.227$, indicating a small to medium effect, suggesting that the difference is meaningful both statistically and practically, thus supporting the hypothesis.

Presence dimension: The hypothesis was supported by users who experienced a significantly higher presence on VR platform than on smartphone platform ($Z = -2.325$, $p = 0.020$). The effect size was $r = -0.231$, indicating a small-to-medium effect and suggesting that the difference was meaningful both statistically and practically.

Immersion dimension: Users did not experience significantly higher immersion on the VR platform than on the smartphone platform ($Z = -0.542$, $p = 0.588$). The effect size was $r = -0.054$, indicating a small effect and suggesting that the practical impact of group differences was weak. However, this hypothesis is not supported.

FS dimension: Users did not experience a significantly higher flow on the VR platform than on the smartphone platform ($Z = -0.744$, $p = 0.457$). The effect size was $r = -0.074$, also indicating a small effect, suggesting a weak practical impact of the difference. However, this hypothesis is not supported. The detailed results are presented in Table 6.

Table 6. Paired Wilcoxon signed-rank test results for four dimensions of VR games and smartphone games

Dimension	VR platform	Smartphone platform	Z	p-value	r	Hypothesis supported
H5: VEC	3.64±0.69	3.39±0.83	-2.277[a]	0.023	-0.227	Supported
H6: EP	3.91±0.49	3.74±0.63	-2.325[a]	0.020	-0.231	Supported
H7: EI	3.89±0.57	3.81±0.67	-0.542[a]	0.588	-0.054	Not supported
H8: FS	4.14±0.51	4.07±0.57	-0.744[a]	0.457	-0.074	Not supported

Values are presented as mean±standard deviation.
[a]Based on negative ranks.

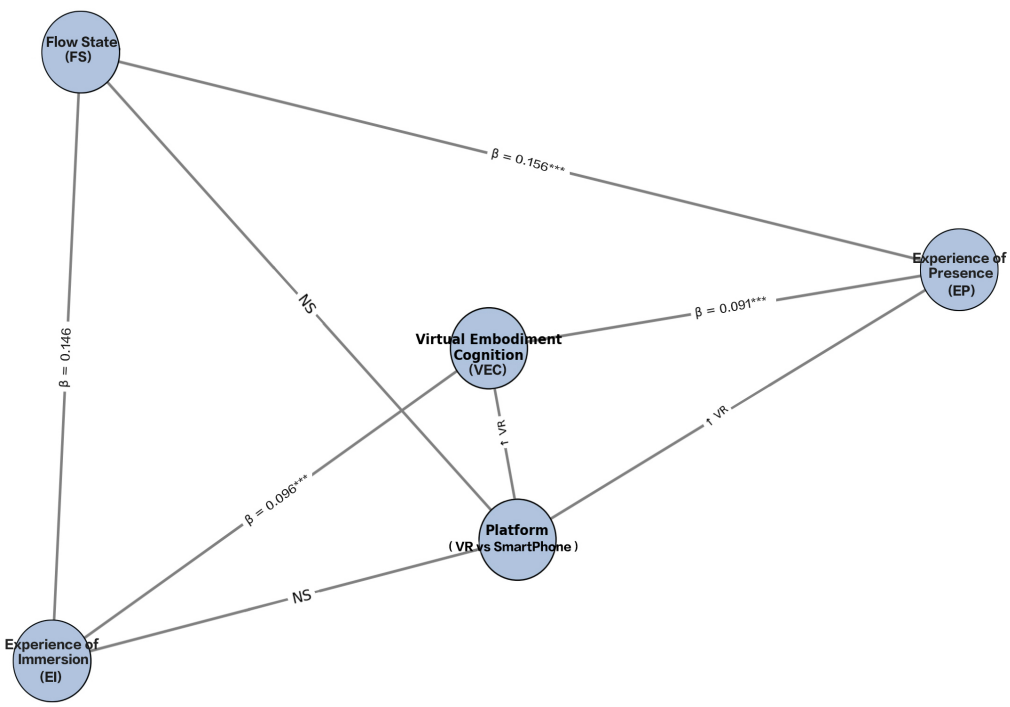


Fig. 3. Theoretical model with moderating effect of platform.

This study constructed and validated a hypothetical model to show how VEC influenced player flow states. The results revealed that VEC significantly and positively predicted both presence and immersion. Presence, in turn, had a significant positive effect on the flow state, whereas the direct impact of immersion on the flow was not statistically significant. Furthermore, a platform comparison analysis showed that the VR platform significantly enhanced users' VEC and sense of presence compared to the smartphone platform, confirming its advantage in fostering contextual awareness (Fig. 3). However, no significant differences were found in the immersion and flow states between the two platforms, suggesting that current VR games are yet to fully surpass traditional mobile platforms in eliciting deeper experiential engagement, indicating room for further improvement.

4. Discussion

4.1 Principal Findings

This study explored the relationships between VEC, EP, EI, and FS, as well as the differences in user experiences across these four dimensions between VR platforms and smartphone platforms. The results indicate that VEC has a significant positive impact on both presence and immersion. Presence had a significant positive impact on the flow state, and immersion had a significant positive impact on flow. A comparison between the VR and smartphone platforms revealed that the VR platform offered a significantly better experience of VEC and presence. VR platforms can provide users with superior experiences through VEC and presence. However, in terms of immersion and flow, the VR platform did not show a clear advantage over the smartphone platform, with only slight differences. This result indicates no significant difference between the VR and smartphone platforms in terms of the immersion and flow state, which is largely inconsistent with the original research expectations.

This study found notable discrepancies in the hypothesized relationship between immersion and flow states, as well as in the comparison of user experiences between VR and smartphone platforms. Specifically, immersion did not significantly influence the flow state in the VR condition, nor did it differ significantly from the smartphone platform, which contradicted the commonly held cognitive assumptions about immersive media. Although some variables (e.g., VEC and sense of presence) showed statistically significant differences, the corresponding effect sizes ranged from small to moderate, indicating that the perceptual impact of the platform on user experience was relatively mild.

These findings suggest that caution should be exercised when overestimating the advantages of VR in current technological structures. The small effect sizes may have resulted from participants' adaptability to multi-platform media, unavoidable differences in sensory inputs across platforms (e.g., FOV and control schemes), and individual differences in sensitivity to immersion. These factors may partially "dilute" the psychological impact of platform differences.

Theoretically, this indicates that the formation of immersion and flow is not solely determined by media type but is also moderated by complex factors such as VBO, multisensory integration, and individual psychological traits. Future research should incorporate additional mediating and moderating variables, such as immersion susceptibility, avatar anthropomorphism, and cognitive engagement, to refine the theoretical pathway model. Nonlinear modeling should also be employed to explore perceptual thresholds and feedback mechanisms.

Practically, these findings have important implications for the industry. On the one hand, they suggest that in certain functional applications—such as education, rehabilitation, or social gaming—mobile platforms can still provide acceptable levels of immersion and flow due to their convenience and accessibility. On the other hand, they highlight the need for developers to shift focus from optimizing single-device performance to designing for cross-platform consistency, thereby promoting a broader adaptation of XR experiences and enabling personalized immersion compensation mechanisms.

4.2 Relationship between Virtual Embodied Cognition and the Experience of Presence

The results of this study demonstrate that VEC has a significant positive effect on presence, thus confirming Hypothesis 1. This underscores the importance of the avatar's body in VR spaces [65]. Because the physical body drives the understanding of the real body in a physical space through embodied sensations and perceptual cognition, users experience their virtual bodies similarly [25]. Synchronization between physical and virtual bodies can effectively enhance a user's sense of presence in a virtual space [29, 66]. This finding is consistent with findings of previous studies.

4.3 Relationship between Virtual Embodied Cognition and the Experience of Immersion

VEC has a significant positive impact on immersion, thus confirming Hypothesis 2. This indicates that an embodied cognitive experience in VR space positively influences immersion acquisition. Immersion is a cognitive experience involving the body's participation in the immersive experience [67]. In interactive presentations, the body captures dynamic data on the user's head, eyes, facial expressions, hands, and limbs using sensors, thereby enabling the physical body to control the virtual body in the game. Embodied perceptions, including vision, hearing, touch, and consciousness, contribute to user interactions in virtual environments [68]. Human body movements, postures, and perceptions of environmental changes affect emotions, comprehension, and creativity, leading to a more immersive experience at the psychological level [48], which aligns with previous research.

4.4 Relationship between the Experience of Presence and Flow State

The results show that presence significantly affects the flow state, thus confirming Hypothesis 3. This suggests that the presence of a virtual body can effectively enhance the flow state of the user. When users perceive spatial presence in a virtual environment through their bodies, they can adjust their posture and movement sensations, thereby experiencing smooth interactions with the virtual environment and aligning their conscious and physical states [43]. The user's cognitive and perceptual systems are "deceived" as believing they are in a virtual scene outside the physical space [54]. This finding is consistent with the findings of existing research.

4.5 Relationship between the Experience of Immersion and Flow State

The results of this study indicate that immersion does not have a significant positive impact on the flow state; therefore, Hypothesis 4 is not supported. This suggests that simply increasing the level of user immersion is not sufficient to trigger a flow state directly. In VR games, although immersion can enhance sensory engagement and contextual involvement, it may not translate into a deep state of flow unless accompanied by cognitive goal alignment, continuous challenge-feedback mechanisms, or a highly focused mental state. This finding implies that the emergence of flow may depend more on the balance between subjective agency and task challenge as well as the user's sense of identification and agency within the virtual environment. Therefore, future game designs should further explore how immersion may indirectly influence flow through other mediating variables, such as an enhanced sense of control, goal clarity, or task competence, to foster sustained player engagement and emotional investment more effectively.

4.6 Comparative Analysis of VR Platform and Smartphone Platform

4.6.1 Comparison in the dimension of virtual embodied cognition

The findings revealed that the VR platform provided a significantly more positive experience of VEC than the smartphone platform, thus confirming Hypothesis 5. This suggests that the unique visual,

auditory, and interactive features of VR platforms surpass flat displays and touch-based interactions of smartphone screens in reproducing the realism of virtual spaces, thus offering users a stronger sense of VEC. Stereoscopic visual effects, spatial audio, and simulated real-world physical interactions in VR significantly enhanced the embodied cognition of virtual avatars. By contrast, the embodied cognitive experience on smartphone platforms primarily relies on emotional projections at the level of consciousness, whereas the VR platform engages both conscious and physical recognition of virtual avatars more effectively [29].

4.6.2 Comparison in the dimension of the experience of presence

The VR platform also demonstrated a significantly more positive experience in the presence dimension than the smartphone platform, thus confirming Hypothesis 6. This indicates that the VR platform establishes a stronger connection between the user's body and the virtual environment through sensory experiences. In contrast, the interaction mode of smartphone platforms predominantly relies on touch-screen operations, with user engagement with game content confined to conscious projection outside the screen and lacking a tangible sense of bodily presence. This finding aligns with the conclusion drawn by Carroll et al. [58] that participants using the VR platform experience a stronger sense of presence, regardless of the game type.

4.6.3 Comparison in the dimension of the experience of immersion

The results show that the VR platform does not exhibit a statistically significant positive difference in the dimension of immersion compared with the smartphone platform, thus failing to confirm Hypothesis 7. Although the data suggested a slightly positive difference in immersion for the VR platform, it did not reach statistical significance. The technical limitations of the Oculus Quest 2 device used in this study, such as its 2K screen resolution, which falls short of achieving a fully realistic visual experience, and its reliance on button-based controller interactions rather than full-hand motion simulation, may have diminished the immersive advantage of the VR platform.

4.6.4 Comparison in the dimension of flow state

Similarly, the VR platform did not demonstrate a statistically significant positive difference in the flow dimension compared to the smartphone platform, thus failing to confirm Hypothesis 8. Although VR platforms enhance user experience in virtual spaces through a stronger sense of presence, current technological conditions are insufficient to enable VR platforms to surpass the flow experiences provided by smartphone platforms. These findings diverge from the conclusions of Pallavicini and Pepe [57], who observed that, under VR and desktop display conditions with comparable game performance, VR platform elicited stronger positive emotions and exhibited more significant advantages in immersion and flow experiences. These differences could be attributed to variations in experimental settings, device types, game content, and participant demographics.

4.7 Concluding Discussion

By integrating VEC theory with the immersion/presence–flow pathway model, this study proposed and empirically validated the role of virtual embodied experience as a psychological mediating mechanism, while further identifying the moderating effect of platform type within this causal chain. The proposed model extends traditional approaches to flow research, both methodologically and theoretically. Regarding flow and platform effects, the use of the Flow State Scale enabled a clearer observation that although platform differences did not significantly impact immersion or flow, VBO and interaction design played crucial roles. Nevertheless, despite the baseline flow levels established in the pre-test, flow remained a subjective experience influenced by individual differences. Future research could incorporate physiological indicators along with psychological measures to improve the accuracy of flow assessment.

Moreover, as the current sample consisted mainly of university students, future studies should consider more diverse populations to enhance the external validity of the findings.

Although this study found a significant positive correlation between VEC experience and immersion, immersion did not show a significant positive correlation with the flow state. Moreover, in the platform comparison, the differences in immersion and flow states between the platforms did not reach a significant level. This outcome may be attributed to several external factors, such as the technical limitations of the current VR hardware, including resolution, latency, and feedback mechanisms. In particular, a lack of full-body haptic feedback may reduce the depth of users' immersive experiences. Additionally, the interaction complexity, character design, and environmental details of the selected game may have influenced the effectiveness of immersion. Although the VR platform has outperformed the smartphone platform in terms of embodied cognition and presence, its advantages in enhancing immersion and flow are less clear. This suggests that increased technological embodiment does not automatically translate into a superior gaming experience. User perceptions of the interaction modes, interface responsiveness, and content familiarity may act as mediators. This finding challenges the assumption of the inherent experiential superiority of VR and warrants further investigation.

At a broader level, the theoretical framework of embodied experience developed in this study can be extended to various media environments. In augmented reality (AR), mixed reality (MR), and even traditional 2D screen-based media, mechanisms of multisensory integration and body ownership illusions (e.g., mirror effects and rubber hand illusion) may similarly enhance user engagement and identification. For example, incorporating sound, motion feedback, or a first-person perspective may evoke a strong virtual embodiment, even in non-immersive settings. This presents new design possibilities for media practitioners: under resource-constrained or lower-tech conditions, sensory illusion strategies can be employed to stimulate user immersion and flow experiences.

Given that this study relied primarily on self-reported measures to assess virtual embodiment, presence, and flow, it is important to acknowledge the limitations of subjective data, particularly their dependence on participants' introspective abilities and linguistic precision. These constraints may hinder an accurate capture of the dynamic and subtle nature of such psychological states. Therefore, it is necessary to reflect critically on the measurement of embodied experiences from a phenomenological perspective. Embodied experience is inherently a subjective state arising from sensory–cognitive integration, characterized by temporality, fluidity, and context dependence. As such, traditional quantitative instruments may fail to fully capture complex structures.

4.8 Limitations

This study has certain limitations. Although this study effectively captured the participants' subjective experiences through self-report scales, such as quantitative measures of flow, immersion, and presence, this method has certain limitations. First, the self-reported data were highly dependent on participants' introspective abilities and self-awareness, that is, their capacity to perceive and articulate their own experiences. Owing to individual differences or contextual factors, participants may report their perceptions and emotional states with varying degrees of accuracy. For example, some individuals may not fully recognize their embodied experiences or immersion in a virtual environment, whereas others may be over reliant on established theoretical concepts to interpret and describe their feelings.

Second, self-report methods typically rely on periodic questionnaires or momentary feedback, which limit our understanding of the temporal dynamics and continuity of embodied experiences. Experiences such as VEC, immersion, and flow are highly fluid and dynamic, and participants' psychological states may rapidly shift within a virtual environment. Current self-reported data may not adequately capture the transient and variable experiential states.

Furthermore, this study used questionnaire-based instruments as the main measurement approach, which might not fully reflect the nuanced effects of multisensory integration on embodied experience. For instance, participants may perceive different levels of ownership of their virtual bodies or varying

degrees of sensory feedback integration, which are subtle distinctions that may not be captured by traditional survey items. Despite these limitations, the proposed mediated path model demonstrates good potential for extension. Future research could apply this model to a wider range of virtual environments, including AR and MR settings, or incorporate additional variables such as user motivation and attentional control, to deepen our understanding of the psychological mechanisms underlying virtual experiences. This path model may also serve as a valuable framework for evaluating and optimizing VR user experiences.

In addition, the study sample consisted mainly of young, technologically literate university students, a relatively homogeneous group that limits the generalizability of the findings. University students tend to be more familiar with digital devices, virtual environments, and interactive gameplay, which may result in stronger levels of reported immersion, presence, and flow than those of the general population. This could have led to an optimistic bias in the results. Future research should include more diverse user groups spanning different age ranges, educational backgrounds, and levels of technological familiarity to enhance the external validity of the findings and their relevance to industrial applications. Moreover, the study employed a single game as the experimental material, which, while useful for controlling content variables and isolating platform differences, limits the generalizability of the conclusions. Future studies should incorporate a broader range of games with different genres and styles to strengthen the applicability of the research outcomes.

4.9 Further Research

The promising findings of this study opens up several avenues for future research and practical applications. First, individual differences should be further explored as potential moderators of the relationship between virtual embodiment and flow. Factors such as embodiment sensitivity, gaming motivation, and immersive tendencies may explain why users respond differently to the same platform experience. Future research could integrate these variables using multigroup SEM or hierarchical linear modeling, providing a more granular understanding of embodied interactions.

Second, future studies should rigorously isolate and control for key technical confounders such as the field of view, input modality, and rendering resolution. Employing a uniform interactive scenario (e.g., the same unity-based prototype) across VR and mobile platforms would allow researchers to assess platform-specific pathways to flow more precisely. In addition, systematically manipulating the avatar's perspective (first-person vs. third-person) and virtual body visibility may reveal further mechanisms underlying presence and ownership.

Third, there is a growing interest in combining physiological and neural measures to complement self-reported data. Tools such as electroencephalogram (EEG), functional near-infrared spectroscopy (fNIRS), and heart rate variability (HRV) could provide insights into the attentional and affective correlates of flow and embodiment. These objective measures would allow researchers to build more integrative neurocognitive models of immersive experiences in virtual environments.

Finally, the theoretical framework developed here, which links virtual body perception, presence, and flow, may inform personalized content recommendations, user engagement scoring systems, and immersive experience Key Performance Indicators (KPIs). These tools can drive innovation in user profiling and adaptive content delivery, particularly in rapidly evolving XR ecosystems.

In summary, this study highlights the need for a more ecologically valid, interdisciplinary, and physiologically grounded understanding of embodied flow experiences, while offering actionable pathways for future research and practical design in immersive media.

5. Conclusion

This study explored the psychological and scientific foundation of VBO by examining the cross-platform effects of multisensory integration on the neural representations of presence and flow states. In

particular, under VR conditions, the interaction between VEC and users' body and consciousness schema significantly influenced their sense of presence and flow. The empirical results indicated that VR platforms significantly enhanced the users' sense of body ownership and immersion through the synchronization of multisensory input and interaction with the virtual body, thereby effectively triggering flow states. In particular, the advantages of the body ownership illusion and simulated interaction in VR media enabled users to enter deeper states of immersion and strengthen their emotional resonance with virtual spaces.

This study further revealed the promoting effects of VEC on presence and immersion. VEC is the integration of external bodily expressions and internal consciousness, enhancing users' sense of being physically present in the virtual environment and reinforcing immersion through emotional resonance. Moreover, this study found that presence had a significant positive effect on the flow state, whereas immersion did not show a significant positive effect. This result suggests that in VR environments, the user's perception of truly "being there" is more conducive to triggering the flow experience. In contrast, although immersion reflected the user's sensory focus and engagement with the virtual scenario, without psychological identification with the virtual character or environment and a sense of real connection, such immersion might not be sufficient to induce a deep state of flow. Therefore, the formation of flow might rely more on the user's sense of "presence" in the virtual space—that is, whether they genuinely felt they existed within the virtual environment, rather than merely being immersed in visual or interactive stimuli. This finding highlights the critical role of presence in virtual experience design and suggests that future VR games and immersive applications should place greater emphasis on enhancing users' sense of connection with their virtual bodies and surroundings to strengthen presence, which might indirectly facilitate the emergence of flow. This indicates that immersion may play a mediating role and needs to interact with other psychological mechanisms to effectively lead players into a state of flow.

Compared with smartphone platforms, this study highlights the significant advantages of VR platforms in virtual embodiment cognition, presence, and interactivity, especially in enhancing users' body perception and interactive experience. However, technological limitations of VR devices, such as insufficient gesture recognition, interaction accuracy, and issues related to headset comfort, restrict the full enhancement of the immersion and flow experience. By contrast, smartphone platforms did not demonstrate the same significant advantages in terms of immersion and flow experiences as those demonstrated on VR platforms, although the high-quality game design on smartphone can trigger flow experiences, particularly in the absence of body involvement.

In summary, this study demonstrates that multisensory integration and the illusion of body ownership in VR play a crucial role in enhancing the presence, immersion, and flow experience, with VR platforms showing distinct advantages in this regard. Future research could explore how technical optimization might overcome the current limitations of VR hardware, thereby enabling more realistic and immersive user experiences and promoting deeper flow states. Notably, the flow was not generated through identical mechanisms across all media. In VR, the neural basis of presence and the construction of agency might lead to a structural reconstruction of flow, offering an opportunity to redefine or expand the flow theory.

Future studies may seek to adapt the flow theory to the media realities of the XR era, investigating whether factors such as identity alignment with avatars, emotional bonding, or cross-platform experience consistency act as new predictors of flow. Such directions could contribute to updating the theory in the context of digital experience psychology. Practically, this study offers valuable guidance for immersive system design and evaluation, particularly in the domains of training, education, and entertainment. In medical simulations and emergency response training, enhancing VBO and flow improves situational awareness and skill retention. In educational gaming, multisensory embodiments foster engagement and narrative transport. Metaverse platforms can benefit from optimizing avatar control fidelity, tactile feedback, and identity-mapping scripts to reinforce user retention and platform loyalty.

Overall, although some of the study's findings may appear predictable, the proposed virtual embodiment–presence/immersion–flow path model fills an important gap in understanding subjective embodied experience mechanisms in VR research. By clarifying the dynamic interrelations among key

psychological variables, this study not only provides a paradigm for variable integration and path modeling in future academic research but also offers embodiment-centered design recommendations for game developers and XR content creators seeking to enhance user immersion and active engagement.

Author's Contributions

Conceptualization, LPF; Methodology, LPF; Resources, LJL; Supervision, LJL; Writing of the original draft, LPF; Writing of the review and editing, NX, LJL; Formal analysis, NX; Data curation, LJL.

Funding

This study was funded by the 2022 Humanities and Social Sciences Research Planning Fund of the Ministry of Education, Research on Chinese-style VR Game Creativity for Spreading Chinese Culture in the Flow Experience Mode (22YJA760040).

Competing Interests

The authors declare that they have no competing interests.

References

[1] J. Katona, "A review of human–computer interaction and virtual reality research fields in cognitive Info-Communications," *Applied Sciences*, vol. 11, no. 6, article no. 2646, 2021. https://doi.org/10.3390/app11062646

[2] E. Navarrete, A. Nehring, S. Schanze, R. Ewerth, and A. Hoppe, "A closer look into recent video-based learning research: a comprehensive review of video characteristics, tools, technologies, and learning effectiveness," *International Journal of Artificial Intelligence in Education*, vol. 35, pp. 1631-1694, 2025. https://doi.org/10.1007/s40593-025-00481-x

[3] G. Bollmer and A. Suddarth, "Embodied parallelism and immersion in virtual reality gaming," *Convergence*, vol. 28, no. 2, pp. 579-594, 2022. https://doi.org/10.1177/13548565211070691

[4] M. Girondini, M. Mariano, G. Stanco, A. Gallace, and L. Zapparoli, "Human bodies in virtual worlds: a systematic review of implicit sense of agency and ownership measured in immersive virtual reality environments," *Frontiers in Human Neuroscience,* vol. 19, article no. 1553574, 2025. https://doi.org/10.3389/fnhum.2025.1553574

[5] W. M. Felton and R. E. Jackson, "Presence: a review," *International Journal of Human–Computer Interaction*, vol. 38, no. 1, pp. 1-18, 2022. https://doi.org/10.1080/10447318.2021.1921368

[6] M. A. Moinnereau, A. A. de Oliveira Jr, and T. H. Falk, "Immersive media experience: a survey of existing methods and tools for human influential factors assessment," *Quality and User Experience*, vol. 7, no. 1, article no. 5, 2022. https://doi.org/10.1007/s41233-022-00052-1

[7] X. Xu, J. Kang, and L. Yan, "Understanding embodied immersion in technology-enabled embodied learning environments," *Journal of Computer Assisted Learning*, vol. 38, no. 1, pp. 103-119, 2022. https://doi.org/10.1111/jcal.12594

[8] D. Shin, "Does augmented reality augment user affordance? The effect of technological characteristics on game behaviour," *Behaviour & Information Technology*, vol. 41, no. 11, pp. 2373-2389, 2022. https://doi.org/10.1080/0144929X.2021.1928286

[9] Y. Bian, C. Zhou, W. Gai, J. Liu, and C. Yang, "The effect of embodied interaction designs on flow experience: examination in VR games," *Virtual Reality*, vol. 27, no. 2, pp. 1549-1565, 2023. https://doi.org/10.1007/s10055-023-00758-3

[10] M. Bujic, M. Salminen, and J. Hamari, "Effects of immersive media on emotion and memory: an experiment comparing article, 360-video, and virtual reality," *International Journal of Human-Computer Studies*, vol. 179, article no. 103118, 2023. https://doi.org/10.1016/j.ijhcs.2023.103118

[11] A. Genay, A. Lecuyer, and M. Hachet, "Being an avatar "for real": a survey on virtual embodiment in augmented reality," *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 12, pp. 5071-5090, 2022. https://doi.org/10.1109/TVCG.2021.3099290

[12] M. Matamala-Gomez, A. Maselli, C. Malighetti, O. Realdon, F. Mantovani, and G. Riva, "Virtual body ownership illusions for mental health: a narrative review," *Journal of Clinical Medicine*, vol. 10, no. 1, article no. 139, 2021. https://doi.org/10.3390/jcm10010139

[13] D. C. Wadle, "The contributions of the bodily senses to body representations in the brain," *Review of Philosophy and Psychology*, vol. 16, no. 3, pp. 855-886, 2025. https://doi.org/10.1007/s13164-024-00756-7

[14] A. Bodzin, R. A. Junior, T. Hammond, and D. Anastasio, "Investigating engagement and flow with a placed-based immersive virtual reality game," *Journal of Science Education and Technology*, vol. 30, no. 3, pp. 347-360, 2021. https://doi.org/10.1007/s10956-020-09870-4

[15] Y. Wang, L. Wang, and K. L. Siau, "Human-centered interaction in virtual worlds: a new era of generative artificial intelligence and metaverse," *International Journal of Human–Computer Interaction*, vol. 41, no. 2, pp. 1459-1501, 2025. https://doi.org/10.1080/10447318.2024.2316376

[16] Y. Gao and C. Spence, "Enhancing presence, immersion, and interaction in multisensory experiences through touch and haptic feedback," *Virtual Worlds*, vol. 4, no. 1, article no. 3, 2025. https://doi.org/10.3390/virtualworlds4010003

[17] Q. Feng, H. Luo, Z. Li, J. Liang, G. Li, and Y. Yi, "Creating an immersive virtual reality game space for multiuser, synchronous co-located collaboration: design considerations and influencing factors," *Applied Sciences*, vol. 14, no. 5, article no. 2167, 2024. https://doi.org/10.3390/app14052167

[18] D. Mougenot and H. Matheson, "Theoretical strategies for an embodied cognitive neuroscience: mechanistic explanations of brain-body-environment systems," *Cognitive Neuroscience*, vol. 15, no. 3-4, pp. 85-97, 2024. https://doi.org/10.1080/17588928.2024.2349546

[19] J. Halak, "Body schema dynamics in Merleau-Ponty," in *Body Schema and Body Image: New Directions*. Oxford, UK: Oxford University Press, 2021, pp. 33-51.

[20] K. Kim, N. Norouzi, G. Jo, D., Bruder, and G. F. Welch, "The augmented reality Internet of Things: opportunities of embodied interactions in transreality," in *Springer Handbook of Augmented Reality*. Cham, Switzerland: Springer, 2023, pp. 797-829. https://doi.org/10.1007/978-3-030-67822-7_32

[21] K. A. Buetler, J. Penalver-Andres, O. Ozen, L. Ferriroli, R. M. Muri, D. Cazzoli, and L. Marchal-Crespo, ""Tricking the brain" using immersive virtual reality: modifying the self-perception over embodied avatar influences motor cortical excitability and action initiation. *Frontiers in Human Neuroscience*, vol. 15, article no. 787487, 2022. https://doi.org/10.3389/fnhum.2021.787487

[22] N. H. Meyer, "Bodily self-consciousness as a framework to link sensory information and self-related components of episodic memory: behavioral, neuroimaging, and clinical evidence," Ph.D. dissertation, École Polytechnique Fédérale de Lausanne, Switzerland, 2023.

[23] C. Peifer, G. Wolters, L. Harmat, J. Heutte, J. Tan, T. Freire, et al., "A scoping review of flow research," *Frontiers in Psychology*, vol. 13, article no. 815665, 2022. https://doi.org/10.3389/fpsyg.2022.815665

[24] J. Chen, C. Ning, and C. Chen, "A systematic literature review of virtual embodied experience," *Psychology & Marketing*, vol. 40, no. 11, pp. 2167-2185, 2023. https://doi.org/10.1002/mar.21876

[25] L. Mejia-Puig and T. Chandrasekera, "The virtual body in a design exercise: a conceptual framework for embodied cognition," *International Journal of Technology and Design Education*, vol. 33, no. 5, pp. 1861-1882, 2023. https://doi.org/10.1007/s10798-022-09793-8

[26] J. Du Toit, "A Merleau-Pontian phenomenology of the virtual: disembodied challenges and embodied prospects," *South African Journal of Philosophy*, vol. 43, no. 4, pp. 307-322, 2024. https://doi.org/10.1080/02580136.2024.2364318

[27] A. Haj-Bolouri, "The Philosopher's Corner: The 'Immersed Flesh': a phenomenological conceptualization of immersive virtual reality embodiment," *ACM SIGMIS Database: the DATABASE for Advances in Information Systems*, vol. 56, no. 2, pp. 13-30, 2025. https://doi.org/10.1145/3733612.3733615

[28] M. Lachmair, M. H. Fischer, and P. Gerjets, "Action-control mappings of interfaces in virtual reality: a study of embodied interaction," *Frontiers in Virtual Reality*, vol. 3, article no. 976849, 2022. https://doi.org/10.3389/frvir.2022.976849

[29] P. Li, F. Qi, and Z. Ye, "Influence of avatar identification on the attraction of virtual reality games: survey study," *JMIR Formative Research*, vol. 8, article no. e56704, 2024. https://doi.org/10.2196/56704

[30] P. Morasso and V. Mohan, "The body schema: neural simulation for covert and overt actions of embodied cognitive agents," *Current Opinion in Physiology*, vol. 19, pp. 219-225, 2021. https://doi.org/10.1016/j.cophys.2020.11.009

[31] J. L. Ponton, V. Ceballos, L. Acosta, A. Rios, E. Monclus, and N. Pelechano, "Fitted avatars: automatic skeleton adjustment for self-avatars in virtual reality," *Virtual Reality*, vol. 27, no. 3, pp. 2541-2560, 2023. https://doi.org/10.1007/s10055-023-00821-z

[32] K. Jung, S. Kim, S. Oh, and S. H. Yoon, "HapMotion: motion-to-tactile framework with wearable haptic devices for immersive VR performance experience," *Virtual Reality*, vol. 28, no. 1, article no. 13, 2024. https://doi.org/10.1007/s10055-023-00910-z

[33] A. Mottelson, A. Muresan, K. Hornbæk, and G. Makransky, "A systematic review and meta-analysis of the effectiveness of body ownership illusions in virtual reality," *ACM Transactions on Computer-Human Interaction*, vol. 30, no. 5, article no. 76, 2023. https://doi.org/10.1145/3590767

[34] D. Dewez, L. Hoyet, A. Lecuyer, and F. Argelaguet Sanz, "Towards "avatar-friendly" 3D manipulation techniques: bridging the gap between sense of embodiment and interaction in virtual reality," in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, Yokohama, Japan, 2021, pp. 1-14. https://doi.org/10.1145/3411764.3445379

[35] S. Bourgeois-Bougrine, N. Bonnardel, J. M. Burkhardt, B. Thornhill-Miller, F. Pahlavan, S. Buisine, J. Guegan, N. Pichot, and T. Lubart, "Immersive virtual environments' impact on individual and collective creativity," *European Psychologist*, vol. 27, no. 3, pp. 237-253, 2022. https://doi.org/10.1027/1016-9040/a000481

[36] M. Lui, K. Y. A. Chong, M. Mullally, and R. McEwen, "Facilitated model-based reasoning in immersive virtual reality: meaning-making and embodied interactions with dynamic processes," *International Journal of Computer-Supported Collaborative Learning*, vol. 18, no. 2, pp. 203-230, 2023. https://doi.org/10.1007/s11412-023-09396-y

[37] M. Rodriguez and E. Kross, "Sensory emotion regulation," *Trends in Cognitive Sciences*, vol. 27, no. 4, pp. 379-390, 2023. https://doi.org/10.1016/j.tics.2023.01.008

[38] M. I. Berkman and E. Akan, "Presence and immersion in virtual reality," in *Encyclopedia of Computer Graphics and Games*. Cham, Switzerland: Springer, 2019, pp. 1-10. https://doi.org/10.1007/978-3-319-08234-9_162-1

[39] K. Zibrek and R. McDonnell, "Social presence and place illusion are affected by photorealism in embodied VR," in *Proceedings of the 12th ACM SIGGRAPH Conference on Motion, Interaction and Games*, Newcastle upon Tyne, United Kingdom, 2019, pp. 1-7. https://doi.org/10.1145/3359566.3360064

[40] D, Paes, J. Irizarry, and D. Pujoni, "An evidence of cognitive benefits from immersive design review: comparing three-dimensional perception and presence between immersive and non-immersive virtual environments," *Automation in Construction*, vol. 130, article no. 103849, 2021. https://doi.org/10.1016/j.autcon.2021.103849

[41] L. Caroux and A. Mougine, "Influence of visual background complexity and task difficulty on action video game players' performance," *Entertainment Computing*, vol. 41, article no. 100471, 2022. https://doi.org/10.1016/j.entcom.2021.100471

[42] A. Kim, M. Kwon, M. Chang, S. Kim, D. Jung, and K. Lee, "The influence of body position on presence when playing a virtual reality game," in *Proceedings of 2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, Osaka, Japan, 2019, pp. 1020-1021. https://doi.org/10.1109/VR.2019.8797870

[43] A. Haans and W. A. IJsselsteijn, "Embodiment and telepresence: toward a comprehensive theoretical framework," *Interacting with Computers*, vol. 24, no. 4, pp. 211-218, 2012. https://doi.org/10.1016/j.intcom.2012.04.010

[44] O. Matthiopoulou, G. Gnecco, M. Sanguineti, D. Mottet, B. Bardy, and A. Camurri, "Towards the automated analysis of expressive gesture qualities in full-body movement: the perceived origin of movement," *Human-centric Computing and Information Sciences*, vol. 14, article no. 54, 2024. https://doi.org/10.22967/HCIS.2024.14.054

[45] R. Silva, D. Brandao, and N. Martins, "The offland exploratory project as a starting point to a literature review on types of immersion in VR," in *Perspectives on Design II*. Cham, Switzerland: Springer, 2021, pp. 339-352. https://doi.org/10.1007/978-3-030-79879-6_25

[46] D. Cirneci, R. Popa, R. L. Ceciu, T. C. Radeanu, A. I. Luca, I. R. Podina, M. Goga, N. Goga, and I. A. Bratosin, "Psychological effects of multisensory VR exposure-based autobiographical memory retrieval," *IEEE Access*, vol. 12, pp. 151825-151833, 2024. https://doi.org/10.1109/ACCESS.2024.3471620

[47] S. Saha, C. Dobbins, A. Gupta, and A. Dey, "Machine learning based classification of presence utilizing psychophysiological signals in immersive virtual environments," *Scientific Reports*, vol. 14, no. 1, article no. 21667, 2024. https://doi.org/10.1038/s41598-024-72376-1

[48] K. Dooley, "A body in the story: exploring body representation, presence and embodiment in virtual reality narratives," in *Virtual Reality Narratives.* Cham, Switzerland: Springer, 2024, pp. 39-67. https://doi.org/10.1007/978-3-031-64965-3_2

[49] M. Csikszentmihalyi, *Flow: The Psychology of Optimal Experience*. New York, NY: Harper & Row, 1990.

[50] S. Agrawal, A. M. D. Simon, S. Bech, K. Bæntsen, and S. Forchhammer, "Defining immersion: literature review and implications for research on audiovisual experiences," *Journal of the Audio Engineering Society*, vol. 68, no. 6, pp. 404-417, 2020. https://doi.org/10.17743/jaes.2020.0039

[51] I. Frochot, S. Elliot, and D. Kreziak, "Digging deep into the experience–flow and immersion patterns in a mountain holiday," *International Journal of Culture, Tourism and Hospitality Research*, vol. 11, no. 1, pp. 81-91, 2017. https://doi.org/10.1108/IJCTHR-09-2015-0115

[52] M. H. Alsamh, A. Hawbani, S. Kumar, and S. H. Alsamhi, "Multisensory metaverse-6G: a new paradigm of commerce and education," *IEEE Access*, vol. 12, pp. 75657-75677, 2024. https://doi.org/10.1109/ACCESS.2024.3392838

[53] M. Slater, M. Usoh, and A. Steed, "Depth of presence in virtual environments," *Presence: Teleoperators & Virtual Environments*, vol. 3, no. 2, pp. 130-144, 1994. https://doi.org/10.1162/pres.1994.3.2.130

[54] E. Patrick, D. Cosgrove, A. Slavkovic, J. A. Rode, T. Verratti, and G. Chiselko, "Using a large projection screen as an alternative to head-mounted displays for virtual environments," in *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*, The Hague, The Netherlands, 2000, pp. 478-485. https://doi.org/10.1145/332040.332479

[55] J. Park, H. Kim, and H. J. Kwon, "The dual dimensions of virtual reality gaming: an empirical study on enjoyment and cybersickness," *Human-centric Computing and Information Sciences*, vol. 14, article no. 62, 2024. https://doi.org/10.22967/HCIS.2024.14.061

[56] A. Alshaer, "The role of sensory experiences in evoking emotional responses in virtual reality," *Journal of Umm Al-Qura University for Engineering and Architecture*, vol. 16, no. 1, pp. 172-184, 2025. https://doi.org/10.1007/s43995-025-00099-z

[57] F. Pallavicini and A. Pepe, "Comparing player experience in video games played in virtual reality or on desktop displays: immersion, flow, and positive emotions," in *Extended abstracts of the Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts*, Barcelona, Spain, 2019, pp. 195-210. https://doi.org/10.1145/3341215.3355736

[58] M. Carroll, E. Osborne, and C. Yildirim, "Effects of VR gaming and game genre on player experience," in *Proceedings of 2019 IEEE Games, Entertainment, Media Conference (GEM)*, New Haven, CT, USA, 2019, pp. 1-6. https://doi.org/10.1109/GEM.2019.8811554

[59] A. K. Jumani, J. Shi, A. A. Laghari, V. V. Estrela, G. A. Sampedro, A. Almadhor, N. Kryvinska, and A. ul Nabi, "Quality of experience that matters in gaming graphics: how to blend image processing and virtual reality," *Electronics*, vol. 13, no. 15, article no. 2998, 2024. https://doi.org/10.3390/electronics13152998

[60] D. Roth and M. E. Latoschik, "Construction of the virtual embodiment questionnaire (VEQ)," *IEEE Transactions on Visualization and Computer Graphics*, vol. 26, no. 12, pp. 3546-3556, 2020. https://doi.org/10.1109/TVCG.2020.3023603

[61] B. G. Witmer and M. J. Singer, "Measuring presence in virtual environments: a presence questionnaire," *Presence*, vol. 7, no. 3, pp. 225-240, 1998. https://doi.org/10.1162/105474698565686

[62] C. Jennett, A. L. Cox, P. Cairns, S. Dhoparee, A. Epps, T. Tijs, and A. Walton, "Measuring and defining the experience of immersion in games," *International Journal of Human-Computer Studies*, vol. 66, no. 9, pp. 641-661, 2008. https://doi.org/10.1016/j.ijhcs.2008.04.004

[63] S. A. Jackson and H. W. Marsh, "Development and validation of a scale to measure optimal experience: the Flow State Scale," *Journal of Sport and Exercise Psychology*, vol. 18, no. 1, pp. 17-35, 1996. https://doi.org/10.1123/jsep.18.1.17

[64] J. Cohen, "Set correlation and contingency tables," *Applied Psychological Measurement,* vol. 12, no. 4, pp. 425-434, 1988. https://doi.org/10.1177/014662168801200410

[65] M. Magalhaes, M. Melo, A. F. Coelho, and M. Bessa, “Affective landscapes: navigating the emotional impact of multisensory stimuli in virtual reality,” *IEEE Access*, vol. 12, pp. 169955-169976, 2024. https://doi.org/10.1109/ACCESS.2024.3499858

[66] R. Kaplan-Rakowski, D. Cockerham, and R. E. Ferdig, “The impact of sound and immersive experience on learners when using virtual reality and tablet: a mixed-method study,” *British Journal of Educational Technology*, vol. 55, no. 4, pp. 1560-1582, 2024. https://doi.org/10.1111/bjet.13417

[67] N. Wenk, J. Penalver-Andres, K. A. Buetler, T. Nef, R. M. Muri, and L. Marchal-Crespo, “Effect of immersive visualization technologies on cognitive load, motivation, usability, and embodiment,” *Virtual Reality*, vol. 27, no. 1, pp. 307-331, 2023. https://doi.org/10.1007/s10055-021-00565-8

[68] P. Banerjee, M. P. Montiel, L. Tomita, O. Means, J. Kutch, and H. Culbertson, “The impact of airflow and multisensory feedback on immersion and cybersickness in a VR surfing simulation,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 5, pp. 2245-2454, 2025. https://doi.org/10.1109/TVCG.2025.3549125